

TE Ranking Comparison — PRCP

Peak TE
n = 670

Rank	Attribute	r
1	q95	+0.624
2	q_mean	+0.612
3	p_mean	+0.588
4	runoff_ratio	+0.441
5	low_prec_freq	-0.405
6	q5	+0.399
7	high_prec_freq	-0.368
8	p_seasonality	-0.366
9	hfd_mean	-0.360
10	aridity	-0.337
11	slope_fdc	+0.288
12	gvf_diff	-0.260
13	high_prec_dur	+0.241
14	gvf_max	+0.224
15	frac_forest	+0.217
16	pet_mean	-0.191
17	stream_elas	-0.177
18	slope_mean	+0.177
19	lai_max	+0.173
20	elev_mean	-0.172
21	high_q_dur	-0.164
22	glim_2nd_class_frac	+0.135
23	glim_1st_class_frac	-0.125
24	frac_snow	-0.112
25	carbonate_rocks_frac	-0.108
26	high_q_freq	-0.105
27	zero_q_freq	-0.101
28	root_depth_50	+0.092
29	soil_conductivity	+0.091
30	geol_porostiy	+0.085
31	clay_frac	-0.075
32	dom_land_cover_frac	+0.068
33	root_depth_99	+0.065
34	sand_frac	+0.064
35	baseflow_index	+0.060
36	low_prec_dur	-0.053
37	geol_permeability	+0.053
38	soil_depth_pelletier	+0.053
39	area_gages2	+0.041
40	low_q_freq	-0.040
41	organic_frac	+0.039
42	area_geospa_fabric	+0.037
43	soil_porosity	-0.030
44	soil_depth_statsgo	+0.029
45	water_frac	-0.012
46	silt_frac	-0.007
47	other_frac	-0.007
48	max_water_content	+0.006
49	lai_diff	-0.006
50	low_q_dur	+0.003

Weighted TE*
n = 670

Rank	Attribute	r
1	q95	+0.440
2	runoff_ratio	+0.399
3	q_mean	+0.388
4	low_prec_freq	-0.380
5	high_prec_freq	-0.363
6	p_mean	+0.316
7	p_seasonality	-0.289
8	aridity	-0.250
9	q5	+0.248
10	slope_mean	+0.235
11	frac_forest	+0.221
12	slope_fdc	+0.212
13	pet_mean	-0.205
14	stream_elas	-0.197
15	frac_snow	+0.156
16	zero_q_freq	-0.151
17	clay_frac	-0.142
18	gvf_max	+0.142
19	gvf_diff	-0.132
20	low_prec_dur	-0.131
21	hfd_mean	-0.128
22	baseflow_index	+0.124
23	high_prec_dur	+0.118
24	glim_1st_class_frac	-0.109
25	dom_land_cover_frac	+0.108
26	silt_frac	+0.107
27	lai_max	+0.102
28	low_q_freq	-0.091
29	glim_2nd_class_frac	+0.089
30	elev_mean	+0.082
31	high_q_freq	-0.082
32	carbonate_rocks_frac	-0.081
33	geol_permeability	-0.078
34	high_q_dur	-0.075
35	area_gages2	+0.056
36	area_geospa_fabric	+0.056
37	sand_frac	+0.035
38	low_q_dur	-0.034
39	geol_porostiy	-0.030
40	soil_conductivity	+0.028
41	organic_frac	+0.028
42	water_frac	+0.027
43	root_depth_50	+0.019
44	soil_porosity	+0.018
45	other_frac	+0.014
46	soil_depth_pelletier	-0.012
47	root_depth_99	-0.011
48	soil_depth_statsgo	-0.010
49	max_water_content	-0.008
50	lai_diff	-0.006

Lag-1 TE
n = 632

Rank	Attribute	r
1	q95	+0.627
2	q_mean	+0.612
3	p_mean	+0.585
4	runoff_ratio	+0.444
5	low_prec_freq	-0.411
6	q5	+0.392
7	high_prec_freq	-0.382
8	p_seasonality	-0.365
9	hfd_mean	-0.360
10	aridity	-0.337
11	slope_fdc	+0.281
12	gvf_diff	-0.254
13	high_prec_dur	+0.244
14	gvf_max	+0.224
15	frac_forest	+0.215
16	pet_mean	-0.207
17	slope_mean	+0.191
18	stream_elas	-0.183
19	lai_max	+0.172
20	elev_mean	-0.158
21	high_q_dur	-0.156
22	glim_2nd_class_frac	+0.146
23	glim_1st_class_frac	-0.131
24	carbonate_rocks_frac	-0.121
25	high_q_freq	-0.109
26	zero_q_freq	-0.108
27	frac_snow	-0.093
28	geol_porostiy	+0.090
29	root_depth_50	+0.084
30	clay_frac	-0.083
31	soil_conductivity	+0.081
32	dom_land_cover_frac	+0.080
33	baseflow_index	+0.064
34	low_prec_dur	-0.054
35	root_depth_99	+0.054
36	sand_frac	+0.053
37	geol_permeability	+0.049
38	area_gages2	+0.047
39	area_geospa_fabric	+0.043
40	low_q_freq	-0.041
41	soil_depth_pelletier	+0.040
42	organic_frac	+0.031
43	soil_porosity	-0.029
44	soil_depth_statsgo	+0.027
45	water_frac	-0.014
46	max_water_content	-0.008
47	low_q_dur	+0.008
48	lai_diff	-0.008
49	other_frac	-0.007
50	silt_frac	+0.002

TE Ranking Comparison — SRAD

Peak TE
n = 653

Rank	Attribute	r
1	q95	+0.603
2	runoff_ratio	+0.574
3	q_mean	+0.517
4	slope_mean	+0.505
5	p_seasonality	-0.480
6	high_prec_freq	-0.480
7	low_prec_freq	-0.404
8	frac_snow	+0.382
9	gvf_diff	-0.372
10	p_mean	+0.361
11	elev_mean	+0.334
12	q5	+0.312
13	stream_elas	-0.289
14	high_prec_dur	+0.288
15	frac_forest	+0.257
16	glim_1st_class_frac	-0.220
17	baseflow_index	+0.204
18	slope_fdc	+0.199
19	aridity	-0.197
20	soil_depth_pelletier	-0.197
21	lai_diff	-0.176
22	other_frac	+0.172
23	glim_2nd_class_frac	+0.172
24	clay_frac	-0.165
25	root_depth_50	-0.151
26	pet_mean	-0.139
27	carbonate_rocks_frac	-0.123
28	zero_q_freq	-0.118
29	max_water_content	-0.094
30	sand_frac	+0.088
31	soil_porosity	-0.086
32	low_q_freq	-0.076
33	geol_porostiy	-0.068
34	dom_land_cover_frac	+0.063
35	soil_depth_statsgo	-0.055
36	low_q_dur	+0.051
37	soil_conductivity	+0.049
38	high_q_dur	+0.047
39	root_depth_99	-0.046
40	low_prec_dur	-0.039
41	high_q_freq	-0.029
42	lai_max	-0.024
43	organic_frac	-0.024
44	hfd_mean	+0.017
45	geol_permeability	+0.012
46	gvf_max	-0.011
47	area_gages2	+0.010
48	water_frac	+0.010
49	area_geospa_fabric	+0.008
50	silt_frac	-0.002

Weighted TE*
n = 653

Rank	Attribute	r
1	q95	+0.606
2	runoff_ratio	+0.561
3	q_mean	+0.533
4	high_prec_freq	-0.458
5	slope_mean	+0.431
6	p_seasonality	-0.426
7	low_prec_freq	-0.414
8	p_mean	+0.379
9	q5	+0.293
10	gvf_diff	-0.291
11	stream_elas	-0.283
12	high_prec_dur	+0.277
13	frac_snow	+0.269
14	slope_fdc	+0.250
15	elev_mean	+0.237
16	pet_mean	-0.225
17	aridity	-0.223
18	frac_forest	+0.222
19	glim_1st_class_frac	-0.199
20	baseflow_index	+0.159
21	soil_depth_pelletier	-0.158
22	glim_2nd_class_frac	+0.154
23	other_frac	+0.148
24	dom_land_cover_frac	+0.139
25	clay_frac	-0.132
26	lai_diff	-0.131
27	zero_q_freq	-0.117
28	root_depth_50	-0.115
29	carbonate_rocks_frac	-0.102
30	max_water_content	-0.098
31	silt_frac	+0.086
32	soil_depth_statsgo	-0.083
33	hfd_mean	-0.075
34	low_q_dur	+0.070
35	root_depth_99	-0.067
36	low_prec_dur	-0.056
37	geol_permeability	-0.044
38	gvf_max	+0.043
39	low_q_freq	-0.040
40	high_q_freq	-0.037
41	geol_porostiy	-0.028
42	organic_frac	-0.026
43	soil_porosity	-0.022
44	water_frac	+0.019
45	sand_frac	+0.018
46	lai_max	+0.012
47	high_q_dur	-0.005
48	area_gages2	+0.002
49	soil_conductivity	+0.002
50	area_geospa_fabric	+0.001

Lag-1 TE
n = 513

Rank	Attribute	r
1	q95	+0.638
2	runoff_ratio	+0.599
3	q_mean	+0.548
4	slope_mean	+0.487
5	high_prec_freq	-0.486
6	p_seasonality	-0.461
7	low_prec_freq	-0.442
8	p_mean	+0.397
9	gvf_diff	-0.326
10	frac_snow	+0.314
11	stream_elas	-0.298
12	q5	+0.284
13	elev_mean	+0.278
14	high_prec_dur	+0.272
15	frac_forest	+0.269
16	aridity	-0.251
17	slope_fdc	+0.248
18	pet_mean	-0.233
19	glim_1st_class_frac	-0.199
20	soil_depth_pelletier	-0.194
21	root_depth_50	-0.176
22	baseflow_index	+0.163
23	glim_2nd_class_frac	+0.152
24	lai_diff	-0.151
25	carbonate_rocks_frac	-0.149
26	clay_frac	-0.140
27	dom_land_cover_frac	+0.139
28	zero_q_freq	-0.129
29	other_frac	+0.127
30	max_water_content	-0.104
31	root_depth_99	-0.096
32	low_prec_dur	-0.073
33	soil_depth_statsgo	-0.072
34	silt_frac	+0.072
35	low_q_dur	+0.063
36	soil_porosity	-0.043
37	geol_permeability	-0.041
38	low_q_freq	-0.040
39	high_q_freq	-0.039
40	organic_frac	-0.038
41	gvf_max	+0.037
42	sand_frac	+0.032
43	hfd_mean	-0.032
44	area_geospa_fabric	-0.029
45	high_q_dur	+0.027
46	area_gages2	-0.025
47	geol_porostiy	-0.022
48	soil_conductivity	+0.017
49	water_frac	-0.012
50	lai_max	+0.007

TE Ranking Comparison — Tair

Peak TE
n = 593

Rank	Attribute	r
1	runoff_ratio	+0.536
2	high_prec_freq	-0.512
3	frac_snow	+0.480
4	q95	+0.455
5	low_prec_freq	-0.448
6	slope_mean	+0.436
7	q_mean	+0.382
8	stream_elas	-0.360
9	elev_mean	+0.342
10	p_seasonality	-0.338
11	q5	+0.252
12	frac_forest	+0.238
13	clay_frac	-0.223
14	baseflow_index	+0.220
15	glim_1st_class_frac	-0.218
16	p_mean	+0.217
17	pet_mean	-0.216
18	slope_fdc	+0.210
19	gvf_diff	-0.200
20	other_frac	+0.185
21	aridity	-0.180
22	soil_depth_pelletier	-0.171
23	glim_2nd_class_frac	+0.164
24	root_depth_50	-0.159
25	zero_q_freq	-0.156
26	hfd_mean	+0.148
27	high_prec_dur	+0.141
28	low_prec_dur	-0.130
29	geol_porostiy	-0.129
30	lai_diff	-0.114
31	low_q_freq	-0.113
32	root_depth_99	-0.088
33	high_q_freq	-0.083
34	sand_frac	+0.079
35	max_water_content	-0.071
36	dom_land_cover_frac	+0.071
37	carbonate_rocks_frac	-0.063
38	area_gages2	+0.058
39	area_geospa_fabric	+0.056
40	geol_permeability	-0.051
41	lai_max	-0.045
42	soil_depth_statsgo	-0.041
43	soil_porosity	-0.040
44	soil_conductivity	+0.034
45	water_frac	+0.033
46	gvf_max	-0.029
47	silt_frac	+0.028
48	organic_frac	+0.007
49	low_q_dur	-0.005
50	high_q_dur	+0.002

Weighted TE*
n = 593

Rank	Attribute	r
1	high_prec_freq	-0.476
2	runoff_ratio	+0.474
3	q95	+0.471
4	low_prec_freq	-0.463
5	q_mean	+0.403
6	p_mean	+0.292
7	stream_elas	-0.279
8	p_seasonality	-0.271
9	slope_fdc	+0.271
10	slope_mean	+0.268
11	pet_mean	-0.251
12	aridity	-0.235
13	frac_snow	+0.229
14	frac_forest	+0.224
15	q5	+0.206
16	low_prec_dur	-0.161
17	zero_q_freq	-0.148
18	elev_mean	+0.146
19	glim_1st_class_frac	-0.140
20	clay_frac	-0.130
21	silt_frac	+0.123
22	dom_land_cover_frac	+0.117
23	gvf_diff	-0.115
24	baseflow_index	+0.114
25	gvf_max	+0.105
26	glim_2nd_class_frac	+0.102
27	soil_depth_pelletier	-0.100
28	geol_permeability	-0.093
29	high_q_freq	-0.088
30	hfd_mean	-0.080
31	high_prec_dur	+0.076
32	other_frac	+0.073
33	lai_max	+0.072
34	high_q_dur	-0.065
35	soil_depth_statsgo	-0.064
36	root_depth_99	-0.060
37	max_water_content	-0.057
38	root_depth_50	-0.052
39	low_q_freq	-0.039
40	water_frac	+0.036
41	carbonate_rocks_frac	-0.030
42	soil_porosity	+0.029
43	low_q_dur	+0.028
44	area_gages2	+0.026
45	area_geospa_fabric	+0.025
46	geol_porostiy	-0.022
47	soil_conductivity	-0.016
48	lai_diff	-0.014
49	organic_frac	+0.002
50	sand_frac	-0.001

Lag-1 TE
n = 358

Rank	Attribute	r
1	frac_snow	+0.465
2	runoff_ratio	+0.465
3	slope_mean	+0.442
4	high_prec_freq	-0.438
5	stream_elas	-0.369
6	low_prec_freq	-0.354
7	elev_mean	+0.340
8	q95	+0.330
9	baseflow_index	+0.329
10	other_frac	+0.275
11	q_mean	+0.275
12	p_seasonality	-0.264
13	glim_1st_class_frac	-0.257
14	clay_frac	-0.252
15	gvf_diff	-0.247
16	hfd_mean	+0.241
17	q5	+0.229
18	geol_porostiy	-0.226
19	root_depth_50	-0.222
20	lai_diff	-0.219
21	glim_2nd_class_frac	+0.207
22	pet_mean	-0.180
23	gvf_max	-0.178
24	lai_max	-0.172
25	slope_fdc	+0.169
26	soil_depth_pelletier	-0.164
27	high_prec_dur	+0.159
28	root_depth_99	-0.156
29	frac_forest	+0.149
30	zero_q_freq	-0.147
31	low_q_freq	-0.136
32	sand_frac	+0.130
33	p_mean	+0.112
34	soil_conductivity	+0.111
35	area_gages2	+0.102
36	area_geospa_fabric	+0.101
37	aridity	-0.097
38	silt_frac	-0.095
39	low_prec_dur	-0.088
40	soil_porosity	-0.084
41	high_q_dur	+0.071
42	water_frac	+0.058
43	high_q_freq	-0.052
44	low_q_dur	+0.044
45	max_water_content	-0.040
46	geol_permeability	+0.037
47	organic_frac	+0.034
48	soil_dept_statsgo	+0.028
49	dom_land_cover_frac	-0.023
50	carbonate_rocks_frac	-0.020

TE Ranking Comparison — VP

Peak TE
n = 577

Rank	Attribute	r
1	frac_snow	+0.480
2	runoff_ratio	+0.479
3	high_prec_freq	-0.454
4	low_prec_freq	-0.397
5	slope_mean	+0.378
6	q95	+0.364
7	elev_mean	+0.344
8	stream_elas	-0.314
9	q_mean	+0.296
10	p_seasonality	-0.249
11	clay_frac	-0.237
12	other_frac	+0.237
13	q5	+0.224
14	hfd_mean	+0.219
15	pet_mean	-0.212
16	slope_fdc	+0.194
17	baseflow_index	+0.191
18	zero_q_freq	-0.177
19	low_prec_dur	-0.176
20	aridity	-0.174
21	frac_forest	+0.169
22	glim_1st_class_frac	-0.168
23	p_mean	+0.151
24	soil_depth_pelletier	-0.136
25	glim_2nd_class_frac	+0.134
26	root_depth_50	-0.133
27	geol_porostiy	-0.127
28	gvf_diff	-0.106
29	low_q_freq	-0.104
30	root_depth_99	-0.090
31	lai_diff	-0.081
32	sand_frac	+0.079
33	dom_land_cover_frac	+0.072
34	soil_conductivity	+0.066
35	high_q_freq	-0.064
36	gvf_max	-0.061
37	lai_max	-0.057
38	silt_frac	+0.057
39	high_prec_dur	+0.049
40	max_water_content	-0.044
41	soil_porosity	-0.043
42	low_q_dur	-0.039
43	area_gages2	+0.037
44	area_geospa_fabric	+0.036
45	water_frac	+0.033
46	carbonate_rocks_frac	-0.031
47	geol_permeability	-0.027
48	high_q_dur	+0.020
49	soil_depth_statsgo	+0.014
50	organic_frac	+0.004

Weighted TE*
n = 577

Rank	Attribute	r
1	low_prec_freq	-0.452
2	high_prec_freq	-0.418
3	runoff_ratio	+0.365
4	q95	+0.338
5	q_mean	+0.292
6	low_prec_dur	-0.287
7	aridity	-0.280
8	p_mean	+0.250
9	slope_fdc	+0.243
10	pet_mean	-0.237
11	zero_q_freq	-0.203
12	stream_elas	-0.202
13	frac_forest	+0.190
14	q5	+0.187
15	frac_snow	+0.184
16	clay_frac	-0.178
17	silt_frac	+0.157
18	slope_mean	+0.147
19	gvf_max	+0.147
20	high_q_freq	-0.130
21	lai_max	+0.123
22	low_q_dur	-0.118
23	high_prec_dur	-0.116
24	high_q_dur	-0.110
25	baseflow_index	+0.105
26	p_seasonality	-0.104
27	geol_permeability	-0.098
28	dom_land_cover_frac	+0.097
29	low_q_freq	-0.095
30	lai_diff	+0.095
31	soil_depth_statsgo	+0.090
32	elev_mean	+0.088
33	glim_1st_class_frac	-0.081
34	max_water_content	+0.078
35	glim_2nd_class_frac	+0.067
36	soil_conductivity	+0.059
37	root_depth_50	+0.057
38	area_gages2	+0.049
39	gvf_diff	+0.048
40	area_geospa_fabric	+0.047
41	soil_porosity	+0.041
42	water_frac	+0.039
43	geol_porostiy	-0.039
44	sand_frac	+0.035
45	organic_frac	+0.033
46	hfd_mean	-0.027
47	root_depth_99	-0.023
48	other_frac	+0.020
49	carbonate_rocks_frac	-0.018
50	soil_depth_pelletier	-0.009

Lag-1 TE
n = 388

Rank	Attribute	r
1	frac_snow	+0.532
2	elev_mean	+0.421
3	runoff_ratio	+0.406
4	slope_mean	+0.403
5	high_prec_freq	-0.373
6	stream_elas	-0.326
7	hfd_mean	+0.320
8	other_frac	+0.294
9	low_prec_freq	-0.279
10	baseflow_index	+0.274
11	q95	+0.227
12	clay_frac	-0.226
13	glim_1st_class_frac	-0.209
14	root_depth_50	-0.208
15	p_seasonality	-0.205
16	zero_q_freq	-0.199
17	geol_porostiy	-0.180
18	q5	+0.179
19	gvf_max	-0.168
20	gvf_diff	-0.167
21	q_mean	+0.165
22	lai_diff	-0.163
23	glim_2nd_class_frac	+0.160
24	root_depth_99	-0.160
25	soil_depth_pelletier	-0.159
26	lai_max	-0.153
27	low_prec_dur	-0.143
28	low_q_freq	-0.141
29	frac_forest	+0.130
30	slope_fdc	+0.127
31	sand_frac	+0.124
32	soil_porosity	-0.116
33	pet_mean	-0.102
34	aridity	-0.095
35	soil_depth_statsgo	+0.078
36	high_q_freq	-0.065
37	soil_conductivity	+0.061
38	geol_permeability	+0.056
39	high_prec_dur	+0.051
40	low_q_dur	-0.047
41	high_q_dur	+0.033
42	organic_frac	-0.030
43	p_mean	+0.029
44	area_gages2	+0.025
45	area_geospa_fabric	+0.023
46	silt_frac	-0.019
47	max_water_content	-0.010
48	water_frac	+0.006
49	dom_land_cover_frac	-0.006
50	carbonate_rocks_frac	+0.003